THE CATHOLIC DIOCESE OF TANGA

TANGA CATHOLIC SECONDARY SCHOOLS UNION (TACASSU)

FORM FOUR JOINT PRE-NATIONAL EXAMINATION 1

033 BIOLOGY

Time: 3 Hours wazaelimu.com **2026**

Instructions

1. This paper consists of sections **A**, **B** and **C** with a total of fifteen questions.
2. Answer all questions in sections A, B, and two questions from section C.
3. All writing should be in blue ink, except for diagrams, which must be drawn in pencil.
4. Cellular phones and any unauthorized materials are not allowed in the examination room.
5. Write your Examination Number on every page of your answer sheets.

SECTION A (16 MARKS)

Answer **all** questions in this section

1. Choose the most correct answer among the given alternatives and write down the answer in the space given.

   (i) During a practical lesson, a student accidentally cuts a finger with a sharp instrument. The wound is bleeding moderately. Which immediate first-aid action is most appropriate?

   A. Wash the wound with dirty water and leave it open
   B. Apply direct pressure with a clean dressing and raise the injured part
   C. Rub soil on the wound to stop bleeding
   D. Tie a very tight string around the injured finger
   E. Pour concentrated disinfectant directly into the wound

   (ii) A student observes an organism that has no chlorophyll, has a cell wall, and obtains nutrients by absorbing substances from dead organic matter. To which kingdom should the organism most appropriately be assigned?

   A. Plantae
   B. Animalia
   C. Fungi
   D. Protoctista
   E. Monera

   (iii) A farmer notices that a young plant bends towards a source of light while growing near a window. Which biological process best explains this response?

   A. Geotropism
   B. Phototropism
   C. Hydrotropism
   D. Chemotropism
   E. Thigmotropism

   (iv) Your friend tells you that when eating eggs, he enjoys eating the egg white but not the egg yolk. Which organelle in a cell is responsible for the synthesis of the part of the egg your friend enjoys?

A. Mitochondrion
B. Ribosome
C. Golgi apparatus
D. Vacuole
E. Chloroplast

(v) The process by which green plants lose water vapor through their stomata is called:
A. Translocation
B. Guttation
C. Transpiration
D. Osmosis
E. Photosynthesis

(vi) Which of the following is a symptom of Scurvy?
A. Swollen abdomen
B. Bleeding gums
C. Bowed legs
D. Poor night vision
E. Skin rashes

(vii) The Seypalm village health center admitted several children patients who were suffering from schistosomiasis. What can be done to prevent the spreading of the disease?
A. Abstaining from sex
B. Wearing masks that cover the mouth and nose
C. Killing pond-water snail
D. Eating warm foods
E. Washing hands after using toilets

(viii) Genetic disorder caused by presence of an extra chromosome number 21, resulting into individuals having a total of 47 chromosomes.
A. Turner's syndrome
B. Polyploidy
C. Sickle cell anaemia
D. Down's syndrome
E. Klinefeleter syndrome

(ix) The main difference between plants and animals is based on …
A. Movement
B. Reproduction
C. Nutrition
D. Excretion
E. Death

(x) Which of the following will not pass through the renal capillary membrane and Bowman's capsule's membrane to enter the nephron of a healthy kidney?
A. Proteins and blood cells
B. Mineral and water
C. Amino acid and urea

D. Glucose and minerals
E. Plasma and water

2. Match items in list A with the correct phrase in list B.

LIST A	LIST B
(i) A student places a clean cloth over a bleeding wound and applies pressure to control the bleeding. (ii) A plant root grows towards a source of water in the soil. (iii) A green plant manufactures glucose using light energy. (iv) A person moves away immediately after touching a very hot object. (v) A microorganism obtains nutrients by absorbing organic matter from a dead organism. (vi) A person with a fracture is prevented from moving the injured limb before receiving medical attention.	A. PhotosynthesisStroke B. First aid C. Hydrotropism D. Reflex action E. Saprophytic nutrition F. Immobilisation G. Transpiration H. Osmosis

SECTION B (54 MARKS)

Answer **all** questions in this section.

3. Human activities such as extraction of raw materials and the production and consumption of goods and services always produce waste. Therefore, production of waste is unavoidable. Furthermore, in order for the produced waste to be well managed and disposed of, the basic principles of waste management must be used. Explain the principles.

4. (a) Differentiate between:
   (i) Natural Immunity and Artificial Immunity
   (ii) Active and passive immunity.
   (b) Briefly explain how the skin acts as a first line of defense against pathogens. Give two points.

5. (a) Of the children who had attended Seypalm health center with their parents in February 2026, two of them were diagnosed with malnutrition conditions; one with Kwashiokor and the other with Marasmus. Briefly explain how was the doctor able to distinguish between the two medical conditions based on their causes and at least two distinctive symptoms of each?
   (b) Name three enzymes found in the pancreatic juice and state the food substance each of the three enzymes acts upon, and the end products of digestion of each food substance due to the digestion by each of the stated enzymes.

6. (a) Draw a well-labeled diagram of a human Red Blood Cell.
   (b) Explain how the Red Blood Cell is adapted to its function of transporting oxygen.

7. (a) Name the respiratory surfaces in the following organisms:
   (i) Fish (ii) Grasshopper (iii) Earthworm (iv) Spider (v) Amoeba (vi) Tadpole (vii) Dog (viii) Crab

(b) Give three characteristics of an efficient respiratory surface.

8. (a) Briefly explain how growth is brought about.
   (b)Why is growth and development important? Give four reasons.

SECTION C (30 MARKS)

Answer any **two** questions from this section.

9. During a biology lesson, students observe that offspring produced by parents of the same species are similar to their parents but are not always exactly identical. The teacher explains that this is related to **variation and evolution**.
   (a) Define the term variation.
   (b) Distinguish between continuous variation and discontinuous variation, giving one example of each.
   (c) Explain three causes of variation among organisms.
   (d) Explain three ways in which variation can contribute to the survival of organisms in a changing environment.

10. "Human activities are the primary cause of environmental degradation in Tanzania." Justify this statement by explaining five human activities that lead to environmental pollution and suggest four measures to be taken for environmental conservation.

11. HIV/AIDS is still one of the scariest diseases in Tanzania. Write a scholarly article about the disease to educate the public on the nature of the disease, the cause of the disease, the way the disease is transmitted, the symptoms and effects of the disease, and the preventive measures and control of the disease.